Cloud Sync Configuration (Supabase)

> 💡 **Tip for returning users (Quick Start):**
>
> If you already have a previously created account and have ready access keys to your database (Server URL and `anon` API key), you don't need to go through the entire installation process. You can completely skip steps 1 to 3 and go directly to **Step 4: Entering Data into the App** to immediately connect your smartphone to the cloud.

By integrating with the Supabase platform, you gain the ability to securely store a full backup of your app data in the cloud. This protects your data from loss in case of damage, loss, or replacement of your phone. This is crucial in farm working conditions, where a phone is exposed to flooding, falling, or crushing in the barn. Synchronization also allows for the seamless restoration of the entire herd history on a new device in just a few seconds.

The app operates on a **local-first** model – this means that all operations are performed instantly on your phone (even without Internet access), and synchronization sends data in the background as soon as you establish a stable connection. The following step-by-step guide will walk you through the free process of setting up, configuring, and securing your own database.

Step 1: Create an account and project in the cloud

The Supabase platform offers a free package (Free Tier), which is more than enough to support the database of a medium to large farm. The transfer and capacity limits will allow you to store information about thousands of animals along with their full treatment history for many years.

1. **Go to the platform:** Open a web browser on your computer or smartphone and visit the homepage: https://supabase.com/
2. **Account Registration:** Click the green **"Start your project"** button in the top right corner. Create a free account. The simplest and fastest method is authorization via a GitHub account. If you don't have one, you can register traditionally using an email address and a strong password. Remember to confirm your registration by clicking the activation link sent to your inbox.
3. **Creating an Organization:** After logging in for the first time, the system will ask you to create a new Organization. This is your private workspace where projects will be stored. You can name it anything (e.g., *MyFarm*) and leave the other settings as default.
4. **New Project:** Inside the newly created organization, click the **"New project"** button. A configuration form will be displayed where you should fill in the following fields:
   - **Name:** Enter a clear name identifying the project, e.g., `herd-app-db`.
   - **Database Password:** Set a very strong password (we recommend using a password generator containing lowercase and uppercase letters, numbers, and special characters). Save it in a safe place – it will only be needed for manual maintenance work on the database.
   - **Region:** Select the server location closest to your place of residence. For users from Poland, the recommended region is **"Central Europe (Frankfurt)"** or **"Warsaw"** (if currently available on the free subscription list). Choosing a nearby server drastically reduces data transfer time and decreases latency during synchronization.
   - **Pricing Plan:** Make sure the **"Free Plan"** is selected. It offers full PostgreSQL database functionality without any hidden fees.
5. **Server Initialization:** Click the **"Create new project"** button. The configuration of a new, dedicated database by Supabase servers is fully automatic and usually takes 2 to 5 minutes. During this time, the server reserves resources and installs necessary software. Wait patiently until the server loading screen disappears.

Step 2: Prepare the database table structure (SQL)

Once your project is successfully launched, you must create an appropriate table with a unique structure tailored to the requirements of the mobile app. Instead of creating a complex network of related tables, the app uses an optimized, modern approach. All basic animal traits are saved in flat columns, while historical data (health records, planned events, and change history) are nested in special `JSONB` type columns. This guarantees lightning-fast performance and no conflicts during data transmission.

1. **Go to the console:** In the left navigation panel on the Supabase dashboard, find and click the **"SQL Editor"** tab (icon with the console symbol `>_`).
2. **New Query:** Click the **"New query"** or **"New Blank Query"** button to open a blank editor worksheet.
3. **Paste the schema:** Copy the following SQL code in its entirety (making sure not to miss any brackets or commas) and paste it directly into the gray editor field:

```
create table public.sync_data (
id text not null,
type text not null,
animal_id text null,
tag text null,
name text null,
dob text null,
status text null,
photo_uri text null,
photo_base64 text null,
mother_tag text null,
father_tag text null,
last_insemination_date text null,
last_insemination_bull text null,
is_archived boolean null,
notes text null,
insemination_pending boolean null,
expected_calving_date text null,
missing_ear_tag boolean null,
medical_records jsonb null default '[]'::jsonb,
events jsonb null default '[]'::jsonb,
animal_changes jsonb null default '[]'::jsonb,
reminder_title text null,
reminder_description text null,
reminder_date text null,
is_cyclic boolean null,
frequency_days integer null,
is_completed boolean null,
updated_at text null,
species text null,
gender text null,
last_estrus_date text null,
preg_cattle integer null,
preg_horse integer null,
preg_pig integer null,
  preg_sheep integer null,
  maturity_cattle integer null,
  maturity_horse integer null,
  maturity_pig integer null,
  maturity_sheep integer null,
  estrus_cycle_cattle integer null default 21,
  estrus_cycle_horse integer null default 21,
  estrus_cycle_pig integer null default 21,
  estrus_cycle_sheep integer null default 17,
  post_calving_duration_days integer null default 30,
  notification_days_before integer null default 3,
  drying_off_days_before integer null default 45,
  preg_goat integer null default 150,
  maturity_goat integer null default 8,
  estrus_cycle_goat integer null default 21,
  registration_date text null,
  constraint sync_data_pkey primary key (id)
) TABLESPACE pg_default;
ALTER TABLE sync_data ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Allow all with anon key"
ON sync_data
FOR ALL TO anon
USING (true)
WITH CHECK (true);
```

💡 Explanation of the table structure:

- `id`: A unique primary key (identifier) that uniquely identifies each record (e.g., `animal_uuid`). This prevents duplicates.
- `medical_records`, `events`, `animal_changes`: Special `jsonb` columns. They store entire lists of events (e.g., the entire history of illnesses and treatments for a given animal) within a single row. This ensures the database doesn't slow down when retrieving a very long treatment history.
- `updated_at`: A key column storing the time of the last modification in milliseconds (`System.currentTimeMillis()`). This lets the app know which data (on the phone or in the cloud) is newer and should overwrite older data during synchronization.

4. **Run the script:** After pasting the code, click the green **"Run"** button located in the bottom right corner or above the editor window.
5. **Verification:** If the operation is successful, the message `"Success. No rows returned"` will appear under the editor. This means your database has been correctly prepared and secured with the **RLS (Row Level Security)** protocol. From now on, only the app equipped with your unique key will be able to connect to it. No unauthorized person will have access to your farm data.

Step 3: Get access keys (API)

To connect the app on your phone with the newly created database, you need two configuration parameters: the server URL and the API access key. Both are automatically generated and unique to your project.

1. **API Settings:** In the left navigation panel of Supabase (at the very bottom), click the gear icon (**Project Settings**), and then select the **"API"** tab from the submenu list.
2. **Copying parameters:** Find the section named **"API Settings"** and copy two parameters from it:
   - **Project URL:** Found in the *URL* field. Copy the entire address (starting with `https://...`) by clicking the **"Copy"** button next to it.
   - **Project API keys:** You will find it in the *Project API keys* section in the row labeled `anon` and `public`. It is a very long string of random characters (JWT Token). Copy it by clicking the **"Copy"** button.

> ⚠️ **Important Security Warning:**
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> In the same section, there is also a key labeled `service_role`. **NEVER** copy it or enter it into the mobile app. This is an administrator key that bypasses RLS security and must remain strictly confidential. Only the secure public `anon` key is intended for use with the app.

Step 4: Entering data into the Mobile App

With the keys ready, you can configure your smartphone to work securely with the cloud.

1. **Open the app:** Launch the app on your phone, go to the main menu, and navigate to the **Settings** tab (the last icon with the gear symbol on the bottom navigation bar).
2. **Expand cloud configuration:** Scroll down, find the section labeled **"Sync Configuration (Supabase)"** (marked with a cloud or info icon), and click it to expand the text fields.
3. **Filling in the fields (Pasting):**
   - In the **Supabase URL** field, paste the previously copied server address (*Project URL*). Make sure the address starts with `https://` and does not contain spaces at the end.
   - In the **Supabase Key** field, paste the previously copied public key (*anon / public key*). Due to the length of the key, we recommend sending it to your phone (e.g., via a messaging app or email) and copying it directly into the Android OS clipboard.
4. **Saving configuration:** The entered data is automatically saved in the secure, encrypted cache of your phone. The configuration section will collapse automatically after clicking outside the text entry area or pressing the back button.

Step 5: Perform the first test sync and understanding the mechanics

Finally, make sure the connection has been established correctly and the cloud database communicates seamlessly with your device.

1. **Force sync:** Staying in the settings tab in the app, look below the configuration fields and click the **"Force Sync"** button.
2. **Process flow:** At this point, the app will pack your entire local Room database (animal profiles, health records, planned calvings, and reminders) into one optimized data package and send it to Supabase, comparing it with the cloud state. Synchronization works according to the following principles:
    - **Last Write Wins:** If the same animal has been modified on two different devices, the app will compare the modification time (`updated_at` expressed in milliseconds). The record with the higher number (i.e., modified later) will be considered final and will overwrite the older record.
    - **No duplication:** Thanks to the unique `id` identifiers, adding notes or diseases on one device will not duplicate the same entries on a second phone.
3. **Final message:**
    - If the configuration is correct, a green message indicating successful synchronization will appear at the bottom of the screen.
    - In case of an error, use the troubleshooting guide below.

Troubleshooting Guide

Symptom / Error	Probable Cause	Solution
Message: **"Sync error. Check connection or API keys"**	1. Typo in the URL or API key. 2. Step 2 not performed (missing table in database). 3. No Internet connection.	1. Make sure the URL starts with `https://` and ends with `.supabase.co` (no slash `/` at the end). 2. Check if a space was accidentally added at the beginning or end when copying the `anon` key. 3. Go to the **SQL Editor** tab in Supabase and ensure the script was successfully run and the `sync_data` table exists.
Data is not saving in the cloud (empty database in Supabase)	Row Level Security (RLS) policies were not enabled or were incorrectly configured.	Ensure that in Step 2 you ran the entire SQL query, including the `ALTER TABLE` and `CREATE POLICY` commands. Without them, the database will reject write attempts from the mobile app for security reasons.
The app freezes or runs very slowly during forced synchronization	Uploading very large animal photos with weak mobile network coverage (e.g., directly in the pasture).	Go to the app settings and make sure you have the **"Wi-Fi Only"** or **"Data Saver"** option enabled. The app will then automatically pause the transmission of heavy image files until you return home and connect to your home Wi-Fi network.

Congratulations! From now on, your farm uses a modern, secure, and completely free data storage system. Your herd is safe, and you have full control over your data.